Lightweight Encrypted Semantic Search via Multi-Channel Modular Signaling

Zahra Arman
Independent Researcher, Plano, TX
zahra.arman.tech@gmail.com

Abstract

We introduce a method for semantic similarity search on encrypted document representations. The system decomposes sentence embeddings into quantized channels, masks them with rejection-sampled cryptographic salt and wave interference, and performs all comparisons in modular arithmetic space. Unlike AES-based approaches where the key holder must decrypt and observe the raw embedding, our system never reconstructs the original signal—even the key holder sees only modular distances. On MSMARCO with 626,906 real passages, the system achieves 98.2% of cosine MRR@10. On two additional retrieval benchmarks (SciFact and NFCorpus), it achieves 90–96%. Across three STS benchmarks it retains 98–100% of baseline. Three embedding models and five languages are evaluated. On identical hardware, the system runs 8× faster than CKKS fully homomorphic encryption and achieves higher retrieval quality than binary, scalar, and product quantization—while being the only encrypted method. A six-part ablation study identifies optimal configuration. Eight security tests pass, with formal proofs under the PRF assumption.

1 Introduction

Semantic search systems embed documents into dense vectors and rank by cosine similarity. These embeddings leak semantic structure: an adversary with database access can cluster documents by topic and infer content [Song & Raghunathan, 2020].

Fully homomorphic encryption (FHE) allows computation on ciphertexts but incurs 39 ms per comparison on a Tesla T4 GPU. We present a system that achieves comparable security at 5 ms—8× faster—using only integer arithmetic.

Threat model. The adversary is an unauthorized observer of the barcode database. The key holder is trusted and performs comparisons, but—unlike AES-based encryption—never observes the raw embedding. In AES, the key holder decrypts, sees the full semantic fingerprint, and computes cosine. In our system, the key holder compensates mask differences in modular space and obtains only a scalar distance. No party ever reconstructs the original embedding.

Security terminology. We use "encrypted" to denote that stored barcodes are computationally indistinguishable from random to any observer without the key, under the PRF assumption. This is a *randomized privacy-preserving encoding*, distinct from reversible block cipher encryption (AES). The transformation is intentionally lossy: the original embedding cannot be recovered even by the key holder.

Distance leakage. A trusted key holder who computes many pairwise distances can, in principle, approximate the embedding geometry via multidimensional scaling (MDS). Empirical testing confirms this: MDS on key-holder distances achieves $\rho = 0.63$ correlation with original geometry. This is inherent to *any* distance-preserving scheme, including CKKS FHE—if distances are useful for search, they encode geometric structure. However, an observer without the key achieves $\rho = 0.002$ ($p = 0.91$), indistinguishable from noise.

Access-pattern leakage (which queries access which barcodes) is outside scope. An optional dummy-query guard is provided in the implementation.

Contributions.

1. A multi-channel modular encryption scheme that preserves 98% of cosine retrieval quality at 626K-document scale.
2. A design rule: modular primes must exceed the quantization bin count. Violation causes 20–50% quality degradation.
3. Same-hardware speed comparison with CKKS FHE: 8× faster.
4. Six-part ablation identifying the contribution of each component.

2 Method

2.1 Projection and Quantization

Project embedding $\mathbf{e} \in \mathbb{R}^d$ onto $K = 200$ PCA components with variance-weighted channels. Quantize each to $\{0, \ldots, B-1\}$ with $B = 50$.

2.2 Cryptographic Masking

For prime p and channel k: $b_{p,k} = (q_k + s_{p,k} + w_k) \bmod p$.

The salt $s_{p,k}$ is drawn via rejection sampling from HMAC-SHA256 bytes, guaranteeing perfect uniformity over $\mathbb{Z}/p\mathbb{Z}$. The wave w_k sums five triangle waves with key-derived parameters, providing defense in depth (ablation shows salt alone is sufficient for security; the wave adds margin).

2.3 Prime Selection Rule

When $p > B$, the map $q \mapsto q \bmod p$ is injective on $\{0, \ldots, B-1\}$: no two distinct signal values produce the same residue. When $p \leq B$, collisions destroy distance information. With $B = 50$, we use $\mathcal{P} = \{53, 59, 61, 67, 71, 73\}$.

This rule was discovered empirically: SciFact retrieval improved from 85% to 96% after enforcing it (Section 3.8).

2.4 Encrypted Comparison

The key holder computes: $D(i,j) = \frac{1}{|\mathcal{P}|} \sum_p \frac{\sum_k \sigma_k^2 |\hat{d}_{p,k}|}{p \sum_k \sigma_k^2}$ where $\hat{d}_{p,k}$ is the mask-adjusted residue difference. The raw quantized signal is never materialized.

3 Experiments

3.1 MSMARCO Passage Retrieval (626,906 documents)

Table 1 reports MRR@10 across corpus sizes and channel counts, all on real MSMARCO passages.

With 200 channels and 500 queries: MiniLM achieves 98.2%, MPNet 99.2%. Quality saturates near 150 channels.

Table 1: MRR@10 as % of cosine on real MSMARCO passages (200 queries each).

Corpus	50ch	100ch	150ch	200ch
5,000	84.9	93.0	100.5	99.0
25,000	77.7	93.3	97.4	98.3
120,000	69.9	88.9	96.9	96.9
626,906	66.5	84.9	97.2	98.6

Table 2: MRR@10 % of cosine on BEIR datasets ($K = 200$, $B = 50$).

Dataset	Domain	Corpus	% Cosine
MSMARCO	Web	626,906	98.2
SciFact	Scientific	5,183	95.7
NFCorpus	Medical	3,633	89.9

3.2 Additional Retrieval Benchmarks

3.3 Multi-Model Evaluation (626K)

BGE performance is lower due to its embedding structure distributing information more uniformly across dimensions, which reduces quantization effectiveness.

3.4 Semantic Textual Similarity

3.5 Multilingual (5 languages, STS17/STS22)

3.6 Comparison with Baselines

Our system achieves the highest quality and is the only encrypted method. “Lightweight” refers to computation (5 ms, integer-only, no GPU for comparison), not storage (1,200 bytes per document).

3.7 Speed: Same Hardware (Colab T4)

3.8 Ablation Study

Key findings from six-part ablation on 24,625 passages:

3.9 Security (8 tests)

4 Formal Security

Three properties proven under the PRF assumption for HMAC-SHA256 (full proofs in supplementary material):

Theorem 1 (Semantic Privacy). *Per-channel privacy is information-theoretic: rejection sampling creates a perfect one-time pad in* $\mathbb{Z}/p\mathbb{Z}$.

Theorem 2 (Pairwise Indistinguishability). *The difference of two independent uniform variables over* $\mathbb{Z}/p\mathbb{Z}$ *is itself uniform, making barcode differences independent of signal differences.*

Theorem 3 (CRT Resistance). *CRT reconstruction recovers* $q_k + m_k$ *exactly, but* m_k *is pseudorandom without the key.*

Table 3: MRR@10 % of cosine across models on 626K real passages.

Model	% Cosine
all-MiniLM-L6-v2	98.2
all-mpnet-base-v2	99.2
BAAI/bge-small-en-v1.5	86.6

Table 4: Spearman ρ % of cosine ($K = 200$, $B = 50$).

Dataset	N	% Cosine
STS-B	1,379	100.1
STS13	1,500	99.3
STS16	1,186	98.4

5 Limitations

A trusted key holder who computes many pairwise distances can approximate embedding geometry via MDS ($\rho = 0.63$ in our test). This is inherent to any distance-preserving scheme, including FHE. Mitigation requires limiting the number of distances a key holder can compute, or using a secure enclave.

Access-pattern leakage is not addressed (standard limitation of searchable encryption). BGE embeddings achieve 87% at scale versus 98% for MiniLM/MPNet; performance is encoder-dependent. NFCorpus (medical domain) yields 90%, below the 98% on web text. Storage is 1,200 bytes/document; "lightweight" refers to computation (integer arithmetic, no GPU), not storage. Independent formal verification of proofs is future work.

The system provides a randomized privacy-preserving encoding, not reversible encryption. The original embedding cannot be recovered, but the distance structure is partially preserved by design.

6 Conclusion

We demonstrated encrypted semantic search at 90–99% of cosine quality across three retrieval benchmarks, three embedding models, and five languages. The system is 8× faster than CKKS FHE, exceeds all tested quantization methods in quality, and is the only encrypted option among them. No party—including the key holder—ever observes the raw embedding.

References

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Table 5: Multilingual % of cosine (paraphrase-multilingual-MiniLM, 250 pairs each).

Language	%	Language	%
Arabic	93.4	French↔En	96.0
Spanish	93.8	Spanish↔En	90.3
Korean	95.1	En↔Arabic	88.2
Chinese	95.4		

Table 6: Comparison on 24,625 MSMARCO passages, 500 queries.

Method	MRR	% Cos	Bytes	Enc
Cosine (float32)	.530	100	1536	No
Binary quant	.514	96.9	48	No
Scalar int8	.524	98.8	384	No
Product quant	.520	97.9	48	No
Ours	.528	99.6	1200	Yes

Table 7: Per-comparison latency on identical hardware.

Method	ms	Encrypted
Cosine	0.025	No
Ours	5.0	Yes
CKKS FHE	38.9	Yes

Table 8: Ablation (MRR % of cosine, 200 queries).

Factor	Setting	%
Channels	50	83.5
	100	94.9
	200	100.3
Bins	5	89.7
	20	101.4
Primes	{3,5,7}	48.0
	{7–23}	100.3
Masking	None	ρ=0.97 leak
	Salt only	ρ=0.03 safe
	Full	ρ=0.01 safe

Table 9: Security evaluation on 300 documents, new configuration.

Attack	Result	Status
IND-CPA	$p = 0.48$	Pass
Statistical	$\rho = 0.10$, $p = 0.08$	Pass
Entropy	100%	Pass
Channel leakage	$\|r\| = 0.30$	Pass
Key recovery	1.0% vs 1.9%	Pass
Chosen-plaintext	$\rho = 0.00$, $p = 0.96$	Pass
Timing	$p = 1.00$	Pass
CRT reconstruction	$\|r\| = 0.01$	Pass